Farhan Islam

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Education

McMaster University Hamilton, ON

Bachelor of Engineering, Electrical Engineering (B.Eng.) *Expected Graduation: Apr 2029*

- **Engineering Award of Excellence**
- Relevant Coursework: Data Structures & Algorithms, Microprocessor Systems, Electronic Devices & Circuits, Signals & Systems, Electromagnetics, Logic Design

Experience

Launch Oversight and Logistics Team Member Sep 2024 – Jan 2025

McMaster Rocketry Club *Hamilton, ON*

- Contributed to rocket **Nimbus**, earning **1st place in the Payload Challenge** and **3rd place in the Basic Category** at Launch Canada 2025
- Coordinated safety validation and hardware–software integration across avionics and telemetry subsystems, supporting **3 successful launches** with **95% safety compliance**
- Diagnosed real-time integration issues during pre-launch testing, reducing subsystem communication failures by **20%** through targeted root cause analysis
- Developed Rust and C++-based telemetry analysis tools, cutting anomaly detection time by **30%** and accelerating post-launch debugging cycles
- Standardized operational workflows, decreasing launch setup time by **15%** and improving procedural reliability

Software Development Intern May 2024 – Aug 2024

IX Technology *Toronto, ON*

- Built a **real-time weather data platform** for 3rd-party clients integrating **RESTful APIs** with asynchronous JavaScript, reducing average data delivery latency by **40%**
- Implemented event-driven frontend architecture supporting **10+** concurrent data streams, reducing UI update latency and improving responsiveness under high-load conditions

Electrical Team Lead Sep 2023 – Apr 2024

Team 9659, FIRST Robotics Competition *Toronto, ON*

- Led cross-functional team of **5+ members** designing robot electrical systems including wiring, power distribution, and motor control, improving **system reliability by 20%**
- Contributed to team earning the **District Championship Rookie All-Star Award** and **FIRST Ontario Provincial Championship Qualifying Award**
- Redesigned power distribution layout to cut energy losses by **25%** and eliminate voltage drops under peak load
- Programmed autonomous and teleoperated control in **C++** with PID tuning and sensor feedback, improving movement accuracy by **15%**

Projects

Accessible Microwave Control System | *Arduino, C++, ESP32, STM32, Bluetooth, WebSocket* |

- Architected a **dual-microcontroller embedded system** delivering **<30ms latency** real-time audio feedback via optimized WebSocket and Bluetooth communication for accessibility applications
- Programmed **ESP32** in **C++** to process ultrasonic sensor data using multi-sensor validation and digital filtering, improving detection reliability in noisy environments by **35%**
- Integrated **STM32** for regulated 5V power distribution across all subsystems, eliminating voltage instability under load

DC Motor Speed Controller with PWM | *Arduino, C++, H-Bridge Driver*

- Designed a **closed-loop motor speed controller** using **PWM** and encoder feedback, achieving **±2 RPM** steady-state accuracy across a 100–3000 RPM operating range
- Applied systematic PID tuning to reduce steady-state error by **90%** and improve step-response stability under varying load, validated through iterative hardware testing
- Implemented H-Bridge bidirectional control with soft-start ramping to prevent inrush current under load

Technical Skills

Programming: C/C++, Python, Rust, MATLAB, Assembly
Embedded Systems: ESP32, STM32, Arduino, I2C/SPI/UART, Real-Time Processing
Power & Control: PWM Control, PID Tuning, Power Distribution
Communication: WebSocket, Bluetooth, Serial Communication
Tools: LTSpice, AutoCAD, Circuit Prototyping
Version Control: Git, GitHub